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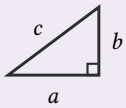
RULE · Pythagorean theorem (Pifagor teoremasi)

$$a^2 + b^2 = c^2$$



Legs a, b meet at the right angle; hypotenuse c is opposite it and always the longest side. Only applies when that right angle is marked or given – SAT figures aren't always drawn to scale.

$a = 9,\ b = 12:\ \ 9^2 + 12^2 = 81 + 144 = 225 \implies c = \sqrt{225} = 15$

RULE · Identify the hypotenuse first

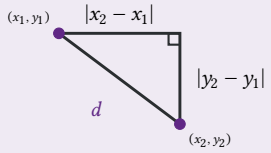
An unknown side isn't always the hypotenuse – it may be a leg. Write the formula with c in the right place.

Hyp. = 13, leg = 5: $\ 5^2 + x^2 = 13^2 \implies x = \sqrt{169 - 25} = 12$

RULES SHEET – 2/2

RULE · Distance formula (masofa formulasi)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



This is just the Pythagorean theorem with the horizontal and vertical legs built from the coordinates.

$$(2,3),(7,15): \quad d = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$$

Each difference is squared, so a negative difference (e.g. $2 - 7 = -5$) still gives a positive result: $(-5)^2 = 25$.

RULE · Common Pythagorean triples

Base triple	Common multiples
3, 4, 5	6, 8, 10 9, 12, 15 12, 16, 20
5, 12, 13	10, 24, 26
8, 15, 17	–
7, 24, 25	–

Not every right triangle's sides form a triple – irrational results like $\sqrt{2}$ or $\sqrt{10}$ are normal too.